

Kubernetes - Kubernetes Introduction

Download & install k3s

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl -sL https://get.k3s.io | sh -s -- --write-kubeconfig
g-mode 644
[INFO] Finding release for channel stable
[INFO] Using v1.32.5+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.32.5+k3s1/sha256sum-amd64.txt
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.32.5+k3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Skipping /usr/local/bin/ctr symlink to k3s, command exists in PATH at /usr/bin/ctr
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s.service -> /etc/systemd/system/k3s.service.
[INFO] systemd: Starting k3s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

View client and server versions

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl version
Client Version: v1.32.5+k3s1
Kustomize Version: v5.5.0
Server Version: v1.32.5+k3s1
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

List single k3s node

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ k3s kubectl get nodes
NAME STATUS ROLES AGE VERSION
ip-172-31-18-113 Ready control-plane,master 99s v1.32.5+k3s1
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

View built-in system pods in kube-system

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pods -A
NAMESPACE NAME READY STATUS RESTARTS AGE
kube-system coredns-697968c856-9lhtj 1/1 Running 0 4m8s
kube-system helm-install-traefik-6775v 0/1 Completed 2 4m8s
kube-system helm-install-traefik-crd-n8l6m 0/1 Completed 0 4m8s
kube-system local-path-provisioner-774c6665dc-p4zzb 1/1 Running 0 4m8s
kube-system metrics-server-6f4c6675d5-n59dr 1/1 Running 0 4m8s
kube-system svclb-traefik-71649094-rm45m 2/2 Running 0 3m38s
kube-system traefik-c98fdf6fb-pjbnz 1/1 Running 0 3m38s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Health of etcd, scheduler, controller-manager

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get componentstatuses
Warning: v1 ComponentStatus is deprecated in v1.19+
NAME STATUS MESSAGE ERROR
etcd-0 Healthy ok
scheduler Healthy ok
controller-manager Healthy ok
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Namespaces (default, kube-system etc)

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get ns
NAME STATUS AGE
default Active 5m41s
kube-node-lease Active 5m41s
kube-public Active 5m41s
kube-system Active 5m42s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```